# Unit Wise Question Bank

**CE/IT/CSE**

# Unit-1

| | |
|---|---|
| Q.1 | What is network topology?Explain different typesof network topology. |
| Q.2 | What is bit and byte stuffing? Explain with example. |
| Q.3 | Explain CRC code generation with example |
| Q.4 | Draw the OSI reference model. Explain the functionality of each layer in brief. |
| Q.5 | Give differences between Connection oriented versus Connection less services |
| Q.6 | Explain functionality of Repeater, HUB, Bridge, Switch, Router and Gateway. |
| Q.7 | Describe Go Back N and Selective Repeat protocol. |
| Q.8 | Discuss the parity checks for error detection in data transfer. |
| Q.9 | What are the issues of stop and wait protocol? How selective repeat protocol resolvesissues of stop andwait protocol? |
| Q.10 | Discuss different types of guided and unguided media used totransmit data in network. |
| Q.11 | What is Hamming Code? Explain how it is useful in error correction. |
| Q.12 | What is the difference between error detection and correction? Explain any One error correction technique with suitable example. |
| Q.13 | Explain TCP/IP Reference Model. Compare TCP/IP with OSI Model |
| Q.14 | Compute CRC for a data sequence 10110111 for which the generator is 110011. |
| Q.15 | What are the reasons for using layered protocols in networks? |
| Q.16 | What is the purpose of cladding in an optical fiber? Discuss its density relative to the core. |
| Q.17 | What is Computer Networks? How is it differing from distributed system? |
| Q.18 | Explain Error Correction Codes at data link layer. |
| Q.19 | Explain different types of Errors? |

| | |
|---|---|
| Q.20 | Explain the simplex protocol for noisy channels. |
| Q.21 | Explain one bit sliding window protocol. Explain one bit sliding window protocol. |
| Q.22 | What kinds of errors can vertical redundancy check determine? What kind of errors it cannot determine? |
| Q.23 | Explain framing and design issues in data link layer. |
| Q.24 | Let the given data bits are 00111001. Compute the hamming code. |
| Q.25 | Given a 6-bit data pattern 100100 and a divisor of 1101, answer the following.<br>i. What will be the dividend at the sender end?<br>ii. Determine the CRC.<br>iii. What will be the transmitted data sequence?<br>iv. What will be remainder at receiver end (assuming no error) ? |

# Unit-2

| | |
|---|---|
| Q.1 | Name and categorizes different multiple access protocols. |
| Q.2 | Compare and explain the pure and slotted ALOHA system. |
| Q.3 | Explain different CSMA protocols |
| Q.4 | What is CSMA/CA? |
| Q.5 | What is CSMA/CD? Explain with Flow chart |
| Q.6 | Explain Manchester encoding for the bit stream 1001110100. |
| Q.7 | How CSMA/CA differs from CSMA/CD. Explain in brief |
| Q.8 | Explain Reservation, Token Passing and POLLING. |
| Q.9 | Explain Binary back off algorithm. |
| Q.10 | Explain controlled Access protocols |
| Q.11 | Compare static vs dynamic channel allocation. |

# Unit-3

| | |
|---|---|
| Q.1 | Compare datagram subnet and virtual circuit subnet. |
| Q.2 | Differentiate adaptive and non adaptive algorithm. |
| Q.3 | Explain shortest path routing algorithm. |
| Q.4 | Explain leaky bucket algorithm. |
| Q.5 | Compare IPv4 & IPv6. |
| Q.6 | What is meant by congestion? List the ways of avoiding congestion. |
| Q.7 | What is flooding? What are its disadvantages? |
| Q.8 | Explain about the IP Header. |
| Q.9 | Explain IP addressing method. |
| Q.10 | Define routing & explain distance vector routing and link state routing. |
| Q.11 | Explain Choke packets, Load shedding, Jitter control. |
| Q.12 | Explain subnetting. |
| Q.13 | Write a note on ICMP. |
| Q.14 | Write a note on RIP. |
| Q.15 | Write a note on OSPF. |
| Q.16 | Differentiate Distance vector routing & Link state routing |
| Q.17 | To solve the examples related to identify netid, hostid, subnet mask. |
| Q.18 | What is extension in IPv6 than IPv4? |
| Q.19 | Explain internetworking. |
| Q.20 | What is firewall? Why it is used? Explain its working. |
| Q.21 | Explain flow based routing |
| Q.22 | Explain shortest pat routing algorithm. |
| Q.23 | Explain IPv6. |
| Q.24 | Draw and explain IPv4 header |
| Q.25 | Compare Classful and classless IP address. |
| Q.26 | Explain ARP protocol. |
| Q.27 | Write a note on working of ICMP protocol. |
| Q.28 | With a real time scenario explain IP addressing and subnetting. |
| Q.29 | Compare distance vector routing and link state routing |
| Q.30 | Explain Flooding. |

# Unit-4

| | |
|---|---|
| Q.1 | Define the socket. List the types of sockets. |
| Q.2 | List the duties of the transport layer and explain each in brief. |
| Q.3 | Compare UDP & TCP. |
| Q.4 | What is meant by congestion? |
| Q.5 | What is function of transport layer? |
| Q.6 | Give the structure of UDP header. |
| Q.7 | Explain TCP header and the working of the TCP protocol. |
| Q.9 | Explain world wide web. |
| Q.10 | Explain HyperText Transfer Protocol. |
| Q.11 | Explain the SMTP. |
| Q.12 | Write a note on DNS. |
| Q.13 | Explain about HTTP. |
| Q.14 | Write short notes on FTP. |
| Q.15 | Explain internet transport protocol. |
| Q.16 | Explain TCP segment header. |
| Q.17 | Difference between absolute domain name and relative domain name. |
| Q.18 | What is domain name system? |
| Q.19 | Explain about MIME. |
| Q.20 | Explain Principle of Congestion Control |
| Q.21 | Draw and explain TCP Header. |
| Q.22 | Draw and explain UDP Header. |
| Q.23 | Difference between leaky bucket and token bucket. |
| Q.24 | Explain with example working of DHCP |
| Q.25 | Explain Leaky Bucket algorithm. |